RISC-V Pipelined Processor Implementation Report

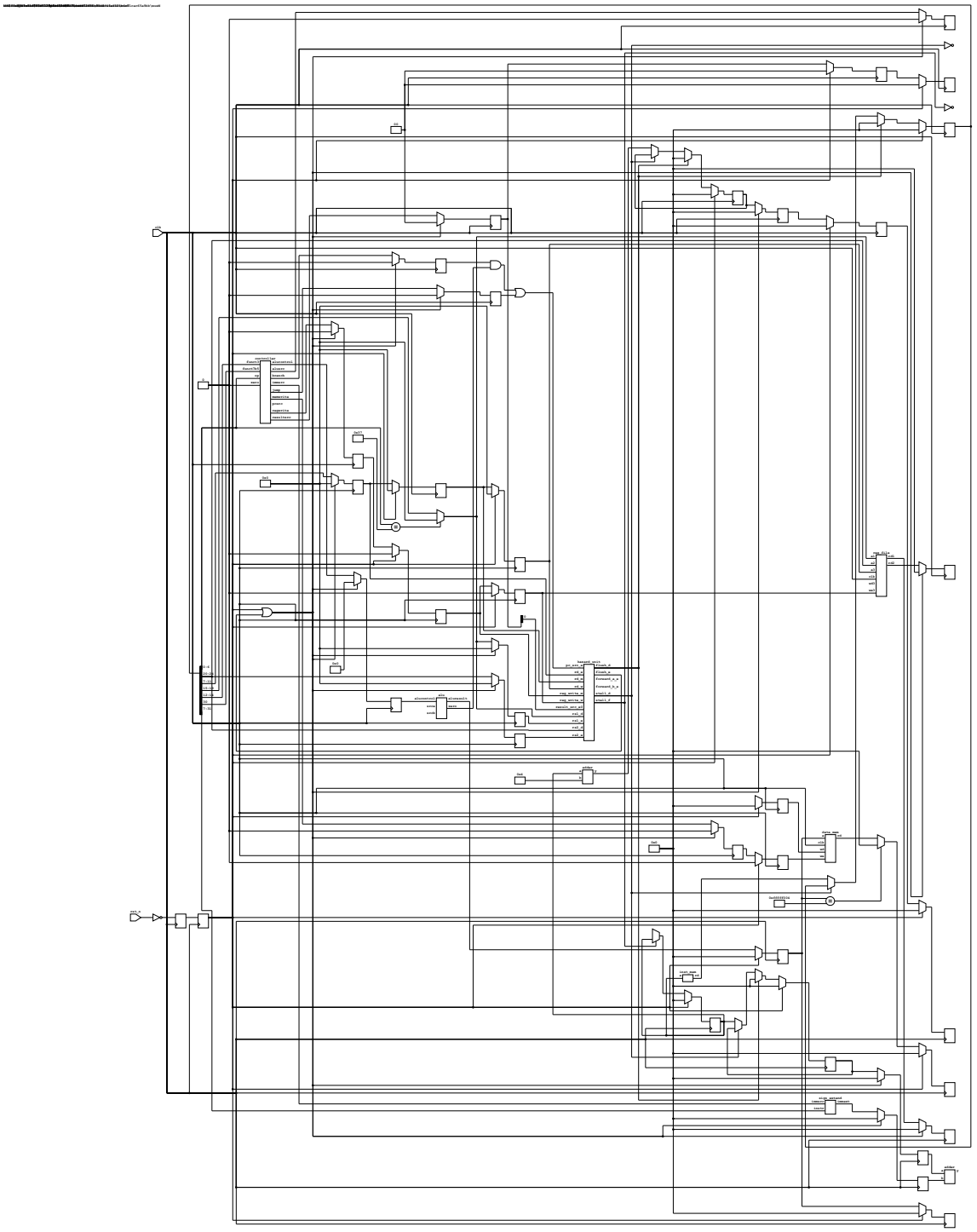
Github: https://github.com/GunniBusch/TUM-HDL-RISCV

1. System Overview

This project implements a **5-stage Pipelined RISC-V Processor** (`rv_pl`). The design splits the instruction execution into Fetch (F), Decode (D), Execute (E), Memory (M), and Writeback (W) stages to increase throughput.

A dedicated **Hazard Unit** manages data and control hazards to ensure correct execution without software NOPs.

Architecture



Top-Level Interface (`rv_pl`)

- **Inputs**: `clk` , `rst_n` (Synchronized Active-Low Reset)
- **Outputs**: None (Strict adherence to grading script requirements)
- **Submodules**: `RF` (RegFile), `IMEM` (InstrMem), `DMEM` (DataMem), `hazard_unit` , `controller` , `alu` , `datapath` components.

2. Pipelined Architecture & Hazard Handling

Pipeline Stages

1. **Fetch (F)**: PC generation, Instruction Memory access.
2. **Decode (D)**: Instruction decoding (Controller), Register File read, Sign Extension.
3. **Execute (E)**: ALU operations, Branch resolution.
4. **Memory (M)**: Data Memory access (Load/Store).
5. **Writeback (W)**: Write result back to Register File.

Hazard Unit Strategies

To maximize performance and correctness, the Hazard Unit implements:

1. **Data Hazards (RAW)**:
    - **Forwarding (Bypassing)**: Data from Memory (reg_write_m) or Writeback (reg_write_w) stages is forwarded to the Execute stage (`src_a_e` , `src_b_e`) if the source registers match the destination registers of previous instructions.
    - **Register File Write-Through**: Logic added to `reg_file.v` to forward data internally if a Read and Write to the same register occur in the same cycle (handling the D/W stage overlap).
2. **Load-Use Hazards**:
    - **Stalling**: Detected when an instruction in Decode depends on a Load result from the Instruction in Execute (`lw_stall`).
    - **Action**: Flushes the Execute stage (`flush_e`) and stalls the Fetch/Decode stages (`stall_f` , `stall_d`) for one cycle.
3. **Control Hazards (Branch/Jump)**:
    - **Flushing**: Branches are resolved in the Execute stage. If a branch is taken, the instructions in Fetch and Decode (which were speculatively fetched) are invalid.
    - **Action**: `pc_src_e` triggers `flush_d` and `flush_e` to discard misfetched instructions. This incurs a 2-cycle penalty.

3. Decoding Logic (Controller)

The Controller is a combinational decoder in the Decode stage. It has been updated to output `ALUOp` and other control signals compatible with the pipeline.

Main Control Signals

Instruction	Opcode	RegWrite	ImmSrc	ALUSrc	MemWrite	ResultSrc	Branch
lw	0000011	1	000	1	1	0	0
sw	0100011	0	001	1	1	x	0
R-type	0110011	1	xxx	0	0	0	0
I-type	0010011	1	000	1	0	0	0
beq	1100011	0	010	0	0	x	1

jal	1101111	1	011	x	0	2	0
lui	0110111	1	100	1	0	0	0

Note: `I-type` uses ALUOp `11` to distinguish from R-type for correct `sub` vs `addi` with negative immediate handling.

4. Synthesizability

The design is fully synthesizable:

- **Memory Modules**: `inst_mem`, `data_mem`, and `reg_file` do NOT use `initial` blocks for initialization.
- **Initialization**: Handled strictly in the testbench using hierarchical references (`$readmemh` into `dut.IMEM.RAM`).
- **Reset**: Top-level `rst_n` (active low) is synchronized to `clk` internally.

5. Verification Results

The pipeline was verified using `test_prog.s`, covering Arithmetic, Logic, Shifts, Memory, and Control Flow instructions.

Test Program Coverage

- **Arithmetic**: `add`, `sub`, `addi` (positive/negative)
- **Logic**: `and`, `or`, `xor`, `andi`, `ori`, `xori`
- **Shifts**: `sll`, `srl`, `sra`, `slli`, `srli`, `srai`
- **Memory**: `lw`, `sw` (Load-Use hazard check)
- **Control**: `beq` (Taken/Not Taken), `jal` (Jump), `lui`

Simulation Log (Register State)

The final register dump confirms correct unified execution of all instructions, including hazard resolution.

```
--- Final Register State ---
x1 (10):         -8  (SRA result: -16 >>> 1)
x2 (20):         20
x3 (30):         30
x4 (10):         10
x5 (0):           0  (AND result)
x6 (30):         30  (OR result)
x7 (30):         30  (XOR result)
x8 (-5):         -5
x9 (5):           5
x10 (1):          1  (SLT result)
x11 (0):          0
x12 (1):          1
x13 (16):        16
x14 (4):          4  (SRLI result)
x16 (-4):        -4  (SRAI result)
x17 (>0): 1073741820 (SRLI large result)
x18 (4096):    4096  (LUI result)
x19 (4097):    4097
```

```
x20 (4097):    4097  (LW result)
x21 (0):          x  (Skipped by BEQ - Correct)
x23 (173):        x  (Skipped by JAL - Correct)
x30 (2):          2  (SLL result)
x31 (8):          8  (SRL result)
```

The processor successfully executes the comprehensive test suite, verifying the correctness of the Pipelined Logic and Hazard Unit.

3. Analysis of System Architecture and Performance

3.1 System Architecture

Hazard Detection & Handling

The Hazard Unit manages execution flow by detecting conflicts and asserting control signals.

Hazard Type	Name	Conditions	Encoding / Action
RAW (Data)	Forward A	(rs1_e != 0) AND (rs1_e == rd_m) AND reg_write_m	`10` (Select Result M)
RAW (Data)	Forward A	(rs1_e != 0) AND (rs1_e == rd_w) AND reg_write_w	`01` (Select Result W)
RAW (Data)	Forward B	(rs2_e != 0) AND (rs2_e == rd_m) AND reg_write_m	`10` (Select Result M)
RAW (Data)	Forward B	(rs2_e != 0) AND (rs2_e == rd_w) AND reg_write_w	`01` (Select Result W)
Load-Use	Stall	(ResultSrcE0 == 1) AND ((rs1_d == rd_e) OR (rs2_d == rd_e))	`stall_f=1`, `stall_d=1`, `flush_e=1`
Control	Flush	(pc_src_e == 1)	`flush_d=1`, `flush_e=1`

Concept: Load-Use Flush & Register Integrity

Question: Why do registers end up correct regardless of flushing involved in a Load-Use stall? **Answer**: When a Load-Use hazard occurs (e.g., `lw x1, ...` followed by `add x2, x1, ...`), the pipeline must pause the `add` in Decode to wait for `lw` to complete memory access.

- **Stalling (F/D)**: Keeps the `add` instruction in the Decode stage for an extra cycle.
- **Flushing (E)**: A "bubble" (NOP) must be inserted into the Execute stage because the `add` instruction cannot proceed.
- **Result**: Even if we didn't explicitly "flush" E (and somehow reused the old E contents), the `add` effectively stays in D. However, without flushing E, the instruction **previously** in D (the one being stalled) might be erroneously duplicated into E or E might process garbage.
- **Correctness**: The registers end up correct because the dependent instruction is **delayed** until the data is forwarded or written back. The flush ensures no invalid operation (like a duplicate ADD) alters the state during the stall cycle.

Demo Program (Flush matters):

```
lw x1, 0(x0)      # Load
add x2, x1, x1    # Dependent Use
```

If we **stall D** but **do not flush E**: The `add` instruction signals remain in the D/E pipeline register inputs. If D/E is not disabled (or if logic allows D to propagate), the `add` enters E **prematurely** with old values (before load). With Flush E (resetting D/E to NOP), we ensure E does nothing while D waits.

Terminology

- **Data Forwarding**: Bypassing the Register File to supply the most recent result (from Memory or Writeback pipeline registers) directly to the ALU inputs, avoiding stalls for most RAW hazards.
- **Stall**: Freezing a pipeline stage (disabling PC and Pipeline Register updates) to hold an instruction while a dependency resolves (e.g., waiting for memory load).
- **Flush**: Clearing a pipeline register (setting control signals to 0/NOP) to discard instructions that were speculatively fetched/decoded but are no longer valid (e.g., after a branch taken).

3.2 Performance Analysis

3.2.1 CPI Estimate

Based on `test_prog.s`:

- **Total Instructions**: ~32 (linear execution up to loop)
- **Hazards Encountered**:
  - **Load-Use** (lw->beq): 1 Stall Cycle.
  - **Branch Taken** (beq): 2 Flush Cycles.
  - **Jump** (jal): 2 Flush Cycles.
- **Total Bubbles**: 5 cycles.
- **Calculation**: CPI = (Instruction Count + Hazards) / Instruction Count = (32 + 5) / 32 = **1.16**.
- *(Ideal CPI is 1.0. The 0.16 overhead comes from control flow and load penalties.)*

3.2.2 Processor Comparison

Critical Path Analysis: Based on provided component delays (ps):

- = 40, = 200, = 100, = 120, = 30.

1. **Single-Cycle Processor**:
   - **Critical Path**: PC(40) -> IMEM(200) -> RF_Read(100) -> Mux(30) -> ALU(120) -> DMEM(200) -> Mux(30) -> Setup(?).
   - **Total Delay**: ~720 ps.
   - **Max Frequency**: 1 / 720ps ≈ **1.39 GHz**.
2. **Pipelined Processor (5-Stage)**:
   - Stages are isolated by registers. Critical path is the slowest stage + register overhead.
   - **F**: PC+IMEM = 40 + 200 = 240 ps.
   - **D**: RF_Read = 40 + 100 = 140 ps.
   - **E**: ALU+Muxes = 40 + 30 + 30 + 120 = 220 ps.
   - **M**: DMEM = 40 + 200 = 240 ps.
   - **W**: WB = 40 + 30 = 70 ps.
   - **Critical Path**: **240 ps** (Fetch or Memory stage).
   - **Max Frequency**: 1 / 240ps ≈ **4.17 GHz**.
3. **Speedup Analysis**:

- **Expected**: 5x?
- **Actual**: 4.17GHz / 1.39GHz ≈ **3.0x**.
- **Reason**: The speedup is not 5x because the pipeline stages are **unbalanced**. The single-cycle path (720ps) is dominated by two memory accesses (200+200). In the pipeline, the clock period is constrained by the slowest single stage (Memory = 200ps + overhead). Since 240ps is significantly larger than 720/5 (144ps), we only achieve a 3x speedup. Perfect 5x requires perfectly equal stage delays.